

source code that runs the front panel can be found on the block diagram. The front panel window contains a toolbar across the top and a 'Controls' palette that you can access by right-clicking anywhere on the front panel. After opening the 'Controls' palette, you can use it to place controls and indicators on the front panel.

Controls and indicators

Controls are the interactive input terminals such as knobs, push buttons, dials and other input devices, while indicators are the interactive output terminals of the VI such as graphs, LEDs and other displays. Controls simulate instrument input devices and supply the block diagram with data. Indicators simulate instrument output devices along with displaying data, which the block diagram acquires.

Every indicator and control has a data type associated with it. To use an example, the Trigger (sec) horizontal

slide is a numeric data type. If you double-click the Trigger control, then LabVIEW jumps to the terminal location on the block diagram. The most commonly used data types are numeric, Boolean value and string.

The block diagram window

When the user creates or opens a new VI, the front panel automatically opens up. To bring up the block diagram, select Window -> Show Block Diagram from the menu bar. Alternatively, you can switch/toggle between the front panel and the block diagram by pressing Ctrl+E.

The components included in a block diagram include functions, sub-VIs, terminals, structures, constants and wires that transfer data among other block diagram objects. A tool is a special operating mode of the mouse cursor, so the operating mode of the cursor corresponds to the icon of the tool selected. The current position of the mouse determines LabVIEW's tool

selection. You can manually choose the tool you need by selecting it on the 'Tools' palette (from the menu bar, select View>Tools Palette). The drag and drop functionality allows you to place objects on the block diagram by simply dragging and dropping them from the 'Functions' palette. This automatically appears when you right-click anywhere on the block diagram workspace with its code containing provisions for functions, constants, structures and some sub VIs.

Functions

Functions are the fundamental operating elements of LabVIEW. They possess neither a front-panel window nor one for the block diagram, but they do come with input and output terminals for passing data in and out—similar to controls and indicators. The functions in the 'Functions' palette are cleverly arranged in groups based on the type of function they perform, thus making the interface extremely user-friendly. ●

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